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TUTORIAL 5 · HEAT BALANCES, TRANSIENT LUMPED · QUICK GUIDE

Coffee with milk: simulation

How it works

1. Open the page with the QR code. Work in pairs on one device.
2. For every question: first note your own answer, then agree in your pair and record your prediction. Only then check it.
3. The page shows results only after your prediction. The explanation is in the card of the question (tab “Experiments”).
4. For question 2 the tutor shares your class's measured values as a link; you take them over with one tap.

QR code of the page
(added after publication)

The three questions of the core path

Cup 1 gets the milk right away, cup 2 only after 60 min.

Start

Both cups stand for 60 min; the milk comes straight from the fridge. Which cup is warmer after 60 min, and by roughly how much? Write down your own answer first, then agree as a pair.

Question 1: time

Write down your own answer first, then agree as a pair: at roughly what time of adding the milk to cup 2 is the difference between the cups largest? Then check: move the time at which the milk is added to cup 2. Why does the difference become smaller again when the milk is added later?

Question 2: experiment

Write down your own answer first, then agree as a pair: do your measured points lie above, below or on the model curves, or sometimes above and sometimes below? Then check: load your class's measured values (link from your tutor) and use the settings from the measurements. Describe where the points lie. Name one possible cause and one measurement you could use to test it.

Afterwards

For fast pairs and for revision: going further V1 to V5 (milk temperature, lid, mug, experiment design, which mug). Again with a prediction for each question.

Your inputs stay in your browser only; nothing is sent. The page also works in German (button “DE ↔ EN”) and offline as a file.